

Schrödinger Bridges with a Reference Drift, and an Analytic Shock-Rotation Drift for a Diamond Airfoil

Abstract

We formulate the Schrödinger-bridge problem as a constrained minimization of kinetic energy, derive its optimality system and Hopf–Cole linearization, and express the solution through log-potentials and the iterative proportional fitting procedure (IPFP). We then extend the formulation to a reference diffusion carrying a macroscopic drift β , obtaining transport–diffusion equations for the Hopf–Cole potentials. Finally we construct an explicit drift β for the compressible flow past a diamond airfoil, built from the oblique-shock geometry, that transports the shock structure between two operating conditions.

Contents

1	Notation	1
2	The Schrödinger problem	2
3	Optimality system and linearization	3
3.1	Euler–Lagrange equations	3
3.2	Hopf–Cole linearization	3
3.3	Log-potentials and IPFP	4
4	Reference process with a drift	4
4.1	Modified problem and optimality system	4
4.2	Transport–diffusion equations	4
4.3	Log-potentials and drifted IPFP	5
4.4	Recovery of the transport map	5
4.5	Baseline transport maps (current implementation, $\beta \equiv 0$)	6
4.6	Convex displacement interpolation of the Mach field	6
5	Analytic reference drift for a diamond airfoil	6
5.1	Geometry	6
5.2	Oblique-shock wave angles	7
5.3	The drift field	7

1 Notation

Vectors lie in $\mathbb{R}^d$ (with $d = 2$ in the application); $\|\cdot\|$ is the Euclidean norm. The symbol γ denotes the diffusion (entropic regularization) coefficient of the bridge; in Section 5 the ratio of specific

heats of the gas is written γ_g to keep the two distinct.

Symbol	Meaning
μ, ν	initial and terminal marginals, $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$
t	time, rescaled to $[0, 1]$
$\rho_t(\mathbf{x})$	density at time t ; $\rho_0 = \mu$, $\rho_1 = \nu$
$b_t(\mathbf{x})$	control (velocity) field
γ	diffusion coefficient of the bridge
$\mathcal{J}$	kinetic-energy functional
$\phi_t(\mathbf{x})$	Lagrange multiplier / value function
η_t, η_t^*	forward / backward Hopf–Cole wave functions, $\rho_t = \eta_t \eta_t^*$
$f_t = \log \eta_t, g_t = \log \eta_t^*$	log-potentials
$\mathcal{P}_\tau$	heat semigroup $e^{\tau \gamma \Delta}$ (Neumann BC)
$\beta_t(\mathbf{x})$	reference drift of the underlying diffusion
$\mathcal{Q}_\tau, \mathcal{Q}_\tau^\dagger$	reference transition operator and its adjoint
α	wedge half-angle of the diamond
γ_g	ratio of specific heats of the gas
M	freestream Mach number; M_0, M_1 its endpoint values
$\theta_{\text{LE}}, \theta_{\text{TE}}$	flow-deflection angles at leading / trailing edge
β_s	oblique-shock wave angle
$\nu(\cdot)$	Prandtl–Meyer function
$\mathbf{a}_{\text{LE}}, \mathbf{a}_{\text{TE}}$	leading- / trailing-edge apex (pivot) points
$\sigma \in \{+1, -1\}$	surface index (upper / lower)
$\kappa \in \{\text{LE}, \text{TE}\}$	shock-family index
$\mathbf{R}$	the rotation matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$
$M_0(\mathbf{x}), M_1(\mathbf{x})$	endpoint Mach fields on the mesh (the quantities interpolated)
T, S	forward ($\mu \rightarrow \nu$) and backward ($\nu \rightarrow \mu$) barycentric transport maps
$\widehat{M}(t, \mathbf{x})$	CDI-reconstructed Mach field at time t
$\text{id}(\mathbf{x}) = \mathbf{x}$	the identity (coordinate) map on the mesh

2 The Schrödinger problem

Let $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ be two probability measures and $\gamma > 0$. The dynamic Schrödinger problem seeks a density–velocity pair $(\rho_t, b_t)_{t \in [0, 1]}$ minimizing the expected kinetic energy

$$\mathcal{J}(\rho, b) = \int_0^1 \int_{\mathbb{R}^d} \frac{1}{2} \|b_t(\mathbf{x})\|^2 \rho_t(\mathbf{x}) d\mathbf{x} dt, \quad (1)$$

subject to the Fokker–Planck constraint and the temporal boundary conditions

$$\partial_t \rho_t + \nabla \cdot (\rho_t b_t) = \gamma \Delta \rho_t, \quad \rho_0 = \mu, \quad \rho_1 = \nu. \quad (2)$$

Equivalently, (ρ, b) is the law of the controlled diffusion $dX_t = b_t(X_t) dt + \sqrt{2\gamma} dW_t$ that is closest, in relative entropy, to the reference Brownian motion of the same intensity; by Girsanov's theorem the relative entropy equals $\mathcal{J}/(2\gamma)$.

3 Optimality system and linearization

3.1 Euler–Lagrange equations

Introduce the multiplier $\phi_t(\mathbf{x})$ for the constraint (2) and form

$$\mathcal{L}(\rho, b, \phi) = \int_0^1 \int_{\mathbb{R}^d} \frac{1}{2} \|b_t\|^2 \rho_t d\mathbf{x} dt + \int_0^1 \int_{\mathbb{R}^d} \phi_t (\partial_t \rho_t + \nabla \cdot (\rho_t b_t) - \gamma \Delta \rho_t) d\mathbf{x} dt. \quad (3)$$

Variation in b .

$$\delta_b \mathcal{L} = \int_0^1 \int_{\mathbb{R}^d} \rho_t b_t \cdot \delta b_t d\mathbf{x} dt + \int_0^1 \int_{\mathbb{R}^d} \phi_t \nabla \cdot (\rho_t \delta b_t) d\mathbf{x} dt. \quad (4)$$

Integrating the second term by parts (no-flux boundary), $\int \phi \nabla \cdot (\rho \delta b) = - \int \rho \nabla \phi \cdot \delta b$, gives $\int_0^1 \int \rho_t (b_t - \nabla \phi_t) \cdot \delta b_t = 0$. Since $\rho_t > 0$ and δb_t is arbitrary,

$$b_t = \nabla \phi_t. \quad (5)$$

Variation in ρ (with $\delta \rho_0 = \delta \rho_1 = 0$). Integrating by parts in time, in the advection term, and twice in the diffusion term (self-adjointness of Δ),

$$\delta_\rho \mathcal{L} = \int_0^1 \int_{\mathbb{R}^d} \left(\frac{1}{2} \|b_t\|^2 - \partial_t \phi_t - b_t \cdot \nabla \phi_t - \gamma \Delta \phi_t \right) \delta \rho_t d\mathbf{x} dt = 0, \quad (6)$$

whence $\partial_t \phi_t + b_t \cdot \nabla \phi_t + \gamma \Delta \phi_t - \frac{1}{2} \|b_t\|^2 = 0$. Substituting (5) yields the Hamilton–Jacobi–Bellman equation

$$\partial_t \phi_t + \frac{1}{2} \|\nabla \phi_t\|^2 + \gamma \Delta \phi_t = 0. \quad (7)$$

The optimality system is (7) coupled to (2) with $b_t = \nabla \phi_t$.

3.2 Hopf–Cole linearization

Define the backward wave function

$$\eta_t^*(\mathbf{x}) = \exp\left(\frac{\phi_t(\mathbf{x})}{2\gamma}\right), \quad \nabla \phi_t = 2\gamma \frac{\nabla \eta_t^*}{\eta_t^*}. \quad (8)$$

A direct computation, using $\Delta \eta^* = \frac{\eta^*}{4\gamma^2} \|\nabla \phi\|^2 + \frac{\eta^*}{2\gamma} \Delta \phi$ and (7), gives the backward heat equation

$$\partial_t \eta_t^* = -\gamma \Delta \eta_t^*. \quad (9)$$

Introduce the forward wave function through the factorization $\rho_t = \eta_t \eta_t^*$. With $b_t = 2\gamma \nabla \log \eta_t^*$ the Fokker–Planck equation (2) becomes

$$\partial_t (\eta \eta^*) + 2\gamma \nabla \cdot (\eta \nabla \eta^*) = \gamma \Delta (\eta \eta^*); \quad (10)$$

expanding and using (9) to cancel the terms in η^* leaves the forward heat equation

$$\partial_t \eta_t = \gamma \Delta \eta_t. \quad (11)$$

The two linear equations are coupled only through the temporal boundary conditions (the Schrödinger system)

$$\eta_0(\mathbf{x}) \eta_0^*(\mathbf{x}) = \mu(\mathbf{x}), \quad \eta_1(\mathbf{x}) \eta_1^*(\mathbf{x}) = \nu(\mathbf{x}). \quad (12)$$

3.3 Log-potentials and IPFP

Set $f_t = \log \eta_t$, $g_t = \log \eta_t^*$, so that

$$\rho_t = e^{f_t + g_t}, \quad b_t = 2\gamma \nabla g_t. \quad (13)$$

Let $\mathcal{P}_\tau = e^{\tau\gamma\Delta}$ be the heat semigroup with homogeneous Neumann boundary conditions. Forward and backward propagation give $\eta_1 = \mathcal{P}_1\eta_0$ and $\eta_0^* = \mathcal{P}_1\eta_1^*$; writing $f := \log \eta_0$ and $g := \log \eta_1^*$, the system (12) reads $f + \log \mathcal{P}_1[e^g] = \log \mu$ and $g + \log \mathcal{P}_1[e^f] = \log \nu$. The iterative proportional fitting procedure (IPFP; Sinkhorn in log form) solves these by alternation:

$$g^{(k+1)} = \log \nu - \log \mathcal{P}_1[e^{f^{(k)}}], \quad f^{(k+1)} = \log \mu - \log \mathcal{P}_1[e^{g^{(k+1)}}]. \quad (14)$$

At convergence the flow is recovered by heat propagation of the endpoint potentials, $\rho_t = \eta_t \eta_t^*$ and $b_t = 2\gamma \nabla \log \eta_t^*$.

4 Reference process with a drift

4.1 Modified problem and optimality system

Let the underlying diffusion carry a prescribed macroscopic drift $\beta_t(\mathbf{x})$,

$$dX_t = \beta_t(X_t) dt + \sqrt{2\gamma} dW_t, \quad (15)$$

and penalize deviation of the control from this reference (Girsanov):

$$\mathcal{J}(\rho, b) = \int_0^1 \int_{\mathbb{R}^d} \frac{1}{2} \|b_t(\mathbf{x}) - \beta_t(\mathbf{x})\|^2 \rho_t(\mathbf{x}) d\mathbf{x} dt, \quad (16)$$

under the same constraint (2). Varying the corresponding Lagrangian in b gives the optimal control

$$b_t = \beta_t + \nabla \phi_t, \quad (17)$$

and varying in ρ , followed by substitution of (17), gives the advected Hamilton–Jacobi–Bellman equation

$$\partial_t \phi_t + \frac{1}{2} \|\nabla \phi_t\|^2 + \beta_t \cdot \nabla \phi_t + \gamma \Delta \phi_t = 0. \quad (18)$$

4.2 Transport–diffusion equations

With the same Hopf–Cole change of variables (8), equation (18) linearizes to the backward transport–diffusion equation

$$\partial_t \eta_t^* + \beta_t \cdot \nabla \eta_t^* + \gamma \Delta \eta_t^* = 0. \quad (19)$$

The forward wave function is characterized as follows.

Proposition 1. *Let $\rho_t = \eta_t \eta_t^*$ solve (2) with optimal drift $b_t = \beta_t + 2\gamma \nabla \eta_t^* / \eta_t^*$, and let η^* satisfy (19). Then*

$$\partial_t \eta_t + \nabla \cdot (\beta_t \eta_t) - \gamma \Delta \eta_t = 0, \quad \text{i.e.} \quad \partial_t \eta_t + \beta_t \cdot \nabla \eta_t + (\nabla \cdot \beta_t) \eta_t - \gamma \Delta \eta_t = 0. \quad (20)$$

Proof. Insert $\rho = \eta\eta^*$ and $\rho b = \eta\eta^*\beta + 2\gamma\eta\nabla\eta^*$ into (2) and expand:

$$\begin{aligned}\partial_t(\eta\eta^*) &= \eta^*\partial_t\eta + \eta\partial_t\eta^*, \\ \nabla\cdot(\eta\eta^*\beta) &= \eta^*\beta\cdot\nabla\eta + \eta\beta\cdot\nabla\eta^* + \eta\eta^*\nabla\cdot\beta, \\ 2\gamma\nabla\cdot(\eta\nabla\eta^*) &= 2\gamma(\nabla\eta\cdot\nabla\eta^* + \eta\Delta\eta^*), \\ \gamma\Delta(\eta\eta^*) &= \gamma(\eta\Delta\eta^* + 2\nabla\eta\cdot\nabla\eta^* + \eta^*\Delta\eta).\end{aligned}$$

Form $\partial_t\rho + \nabla\cdot(\rho b) - \gamma\Delta\rho = 0$ and eliminate $\partial_t\eta^*$ with (19), i.e. $\eta\partial_t\eta^* = -\eta\beta\cdot\nabla\eta^* - \gamma\eta\Delta\eta^*$. The terms $\eta\beta\cdot\nabla\eta^*$ cancel; the terms $\nabla\eta\cdot\nabla\eta^*$ cancel $(+2\gamma - 2\gamma)$; and the terms $\eta\Delta\eta^*$ cancel $(-\gamma + 2\gamma - \gamma)$. What remains is $\eta^*\partial_t\eta + \eta^*\beta\cdot\nabla\eta + \eta\eta^*\nabla\cdot\beta - \gamma\eta^*\Delta\eta = 0$; dividing by $\eta^* > 0$ gives (20). $\square$

Remark 1. The backward equation (19) carries the advection in non-conservative form $\beta\cdot\nabla\eta^*$ (the generator $\mathcal{L} = \beta\cdot\nabla + \gamma\Delta$), whereas the forward equation (20) carries it in conservative form $\nabla\cdot(\beta\eta)$ (the adjoint $\mathcal{L}^*$). The reaction term $(\nabla\cdot\beta)\eta$ is therefore present whenever $\nabla\cdot\beta \neq 0$ and vanishes only for a divergence-free reference field.

4.3 Log-potentials and drifted IPFP

With $f_t = \log\eta_t$ and $g_t = \log\eta_t^*$, equations (19)–(20) become the coupled Hamilton–Jacobi equations

$$\partial_t g_t + \beta_t\cdot\nabla g_t + \gamma\Delta g_t + \gamma\|\nabla g_t\|^2 = 0, \quad (21)$$

$$\partial_t f_t + \beta_t\cdot\nabla f_t + (\nabla\cdot\beta_t) - \gamma\Delta f_t - \gamma\|\nabla f_t\|^2 = 0. \quad (22)$$

Let $\mathcal{Q}_\tau = e^{\tau\mathcal{L}}$ be the transition operator of (15) (the non-conservative backward semigroup) and $\mathcal{Q}_\tau^\dagger = e^{\tau\mathcal{L}^*}$ its adjoint (the conservative forward semigroup). Propagation of the endpoint potentials gives $\eta_0^* = \mathcal{Q}_1\eta_1^*$ and $\eta_1 = \mathcal{Q}_1^\dagger\eta_0$, and the IPFP reads

$$f^{(k+1)} = \log\mu - \log\mathcal{Q}_1[e^{g^{(k)}}], \quad g^{(k+1)} = \log\nu - \log\mathcal{Q}_1^\dagger[e^{f^{(k+1)}}]. \quad (23)$$

When $\beta \equiv 0$ both operators reduce to the heat semigroup $\mathcal{P}_1$ and (23) returns (14).

4.4 Recovery of the transport map

The bridge yields the forward map T (the conditional mean under the coupling),

$$T(\mathbf{x}) = \frac{\mathcal{Q}_1[\text{id} \cdot e^g](\mathbf{x})}{\mathcal{Q}_1[e^g](\mathbf{x})} - \beta_{\text{bias}}(\mathbf{x}), \quad \beta_{\text{bias}} = \frac{\mathcal{Q}_1[\text{id}]}{\mathcal{Q}_1[1]} - \text{id}, \quad (24)$$

with an analogous backward map from e^f ; β_{bias} removes the identity leak of the Neumann semigroup. Evaluated in the log domain, with a shift c such that $\text{id} - c > 0$,

$$T_j(\mathbf{x}) = c_j + \exp\left(\log\mathcal{Q}_1[e^{\log(\text{id}_j - c_j) + g}](\mathbf{x}) - \log\mathcal{Q}_1[e^g](\mathbf{x})\right) - (\beta_{\text{bias}})_j, \quad (25)$$

which is well defined uniformly in γ .

4.5 Baseline transport maps (current implementation, $\beta \equiv 0$)

The pipeline in its present form uses the driftless bridge, so $\mathcal{Q}_1 = \mathcal{Q}_1^\dagger = \mathcal{P}_1$ is the pure heat semigroup and (24) specialises to a *pair* of barycentric maps: the forward map T (conditional mean of the target position given the source, built from g) and the backward map S (conditional mean of the source given the target, built from f),

$$T(\mathbf{x}) = \frac{\mathcal{P}_1[\text{id } e^g](\mathbf{x})}{\mathcal{P}_1[e^g](\mathbf{x})} - \beta_{\text{bias}}(\mathbf{x}), \quad S(\mathbf{x}) = \frac{\mathcal{P}_1[\text{id } e^f](\mathbf{x})}{\mathcal{P}_1[e^f](\mathbf{x})} - \beta_{\text{bias}}(\mathbf{x}), \quad (26)$$

sharing the single Neumann identity-leak correction $\beta_{\text{bias}} = \mathcal{P}_1[\text{id}]/\mathcal{P}_1[1] - \text{id}$. Each of the five distinct heat solves in (26) is evaluated through the stable log-domain operator $\log \mathcal{P}_1[e^{(\cdot)}]$, with the positivity shift $\text{id} \mapsto \text{id} - c$ ($c = \min_{\mathbf{x}} \text{id} - 1$) applied componentwise so that $\log(\text{id} - c)$ is finite; this is exactly the form (24) with $\mathcal{Q}_1$ replaced by $\mathcal{P}_1$. Both maps reduce to the identity far from the transported feature (where β_{bias} cancels the leak) and to the entropic Monge map on its support.

4.6 Convex displacement interpolation of the Mach field

The maps T, S act on the shock-feature marginals; the physical field is recovered by a gradient-free *convex displacement interpolation* (CDI). For $t \in (0, 1)$,

$$\widehat{M}(t, \mathbf{x}) = (1 - t) M_0((1 - t) \mathbf{x} + t S(\mathbf{x})) + t M_1(t \mathbf{x} + (1 - t) T(\mathbf{x})), \quad (27)$$

where M_0, M_1 are the endpoint Mach fields, sampled at the displaced pre-images by linear interpolation on the mesh (nearest-neighbour fallback inside the solid body). The construction is exact at the endpoints, $\widehat{M}(0, \cdot) = M_0$ and $\widehat{M}(1, \cdot) = M_1$, and needs no map inversion: the forward pre-image $t \mathbf{x} + (1 - t) T(\mathbf{x})$ pulls back M_1 , the backward pre-image $(1 - t) \mathbf{x} + t S(\mathbf{x})$ pulls back M_0 , and the two are blended with the convex weights $(1 - t, t)$. This is the object the analytic drift of Section 5 is designed to improve: by pre-transporting the shocks, β leaves T, S close to the identity and keeps (27) well-conditioned at small γ .

5 Analytic reference drift for a diamond airfoil

The endpoints μ, ν are the shock-feature marginals of the Mach fields at freestream Mach numbers M_0 and M_1 . We construct β_t so that it transports the shock structure between the two conditions; the bridge of Section 4 supplies the remaining correction.

5.1 Geometry

The diamond is a symmetric rhombus of chord c , half-chord $h_c = c/2$, total height $H = c \tan \alpha$, half-height $h_t = H/2$, centred at (c_x, c_y) , with vertices

$$\underbrace{(c_x - h_c, c_y)}_{\text{LE}}, \quad \underbrace{(c_x, c_y + h_t)}_{\text{upper shoulder}}, \quad \underbrace{(c_x + h_c, c_y)}_{\text{TE}}, \quad \underbrace{(c_x, c_y - h_t)}_{\text{lower shoulder}}.$$

From the vertices, independent of M ,

$$\mathbf{a}_{\text{LE}} = (c_x - h_c, c_y), \quad \mathbf{a}_{\text{TE}} = (c_x + h_c, c_y), \quad \theta_{\text{LE}} = \theta_{\text{TE}} = \arctan \frac{h_t}{h_c} = \alpha, \quad (28)$$

and the shoulder produces an expansion through the angle $\Delta\theta_{\text{sh}} = 2\alpha$.

5.2 Oblique-shock wave angles

For an attached oblique shock produced by deflection θ at Mach M in a gas of ratio of specific heats γ_g , the wave angle β_s solves the θ - β - M relation

$$\tan \theta = 2 \cot \beta_s \frac{M^2 \sin^2 \beta_s - 1}{M^2(\gamma_g + \cos 2\beta_s) + 2}; \quad (29)$$

denote by $\beta_s(\theta, M)$ its weak-shock root (the smaller root, exceeding the Mach angle $\arcsin(1/M)$).

Assumption 1. $\alpha < \theta_{\max}(M) := \max_{\beta_s} \theta(\beta_s, M)$ for all $M \in [M_0, M_1]$, so both shock families are attached, and each shock is modelled as a straight ray from its edge.

Leading-edge family. The LE shock deflects the freestream by $\theta_{\text{LE}} = \alpha$ at the freestream Mach M :

$$\beta_s^{\text{LE}}(M) = \beta_s(\alpha, M). \quad (30)$$

Trailing-edge family. The TE shock processes the flow that has crossed the LE shock and expanded around the shoulder. With $\beta = \beta_s^{\text{LE}}(M)$ and $M_{n1} = M \sin \beta$, the Mach behind the LE shock is

$$M_{n2}^2 = \frac{1 + \frac{\gamma_g - 1}{2} M_{n1}^2}{\gamma_g M_{n1}^2 - \frac{\gamma_g - 1}{2}}, \quad M_a = \frac{M_{n2}}{\sin(\beta - \alpha)}. \quad (31)$$

The Prandtl–Meyer function

$$\nu(M) = \sqrt{\frac{\gamma_g + 1}{\gamma_g - 1}} \arctan \sqrt{\frac{\gamma_g - 1}{\gamma_g + 1} (M^2 - 1)} - \arctan \sqrt{M^2 - 1} \quad (32)$$

gives the post-shoulder Mach M_b through $\nu(M_b) = \nu(M_a) + 2\alpha$, and the TE wave angle follows with deflection $\theta_{\text{TE}} = \alpha$ at Mach M_b :

$$\beta_s^{\text{TE}}(M) = \beta_s(\alpha, M_b(M)). \quad (33)$$

Both wave angles are evaluated at every interpolation node $M_i = M(t_i)$ rather than interpolated between the endpoints.

5.3 The drift field

Fix the schedule $M(t) = (1 - t)M_0 + tM_1$ and set $\beta_s^\kappa(t) := \beta_s^\kappa(M(t))$ for $\kappa \in \{\text{LE}, \text{TE}\}$. Define the ray-angle magnitudes

$$\psi_{\text{LE}}(t) = \beta_s^{\text{LE}}(t), \quad \psi_{\text{TE}}(t) = \beta_s^{\text{TE}}(t) - \alpha, \quad (34)$$

the subtraction accounting for the local upstream flow along the rear face; the lower ray angle is the negative of the upper, so the two surfaces are mirror images. The unit ray directions and angular rates are, for $\sigma \in \{+1, -1\}$,

$$\hat{\mathbf{d}}_{\kappa, \sigma}(t) = (\cos(\sigma\psi_\kappa), \sin(\sigma\psi_\kappa)), \quad \omega_\kappa(t) = \dot{\psi}_\kappa(t) = \left. \frac{d\beta_s^\kappa}{dM} \right|_{M(t)} (M_1 - M_0). \quad (35)$$

Since β_s decreases with M , $\omega_\kappa < 0$: each shock rotates toward the body. A point at $\mathbf{x}$, riding on ray (κ, σ) as it rotates about $\mathbf{a}_\kappa$, has velocity $\sigma \omega_\kappa \mathbf{R}(\mathbf{x} - \mathbf{a}_\kappa)$, using $\mathbf{R}(\cos \varphi, \sin \varphi) = (-\sin \varphi, \cos \varphi)$.

Localizing to the shock by a Gaussian band of width σ_w in the perpendicular distance, gated downstream, define

$$s_{\kappa,\sigma} = (\mathbf{x} - \mathbf{a}_\kappa) \cdot \hat{\mathbf{d}}_{\kappa,\sigma}, \quad \text{dist}_{\kappa,\sigma} = \|(\mathbf{x} - \mathbf{a}_\kappa) - s_{\kappa,\sigma} \hat{\mathbf{d}}_{\kappa,\sigma}\|, \quad w_{\kappa,\sigma} = e^{-\text{dist}_{\kappa,\sigma}^2 / \sigma_w^2} \mathbf{1}[s_{\kappa,\sigma} > 0]. \quad (36)$$

The reference drift is

$$\boxed{\boldsymbol{\beta}(t, \mathbf{x}) = \sum_{\kappa \in \{\text{LE}, \text{TE}\}} \sum_{\sigma \in \{+1, -1\}} \sigma \omega_\kappa(t) w_{\kappa,\sigma}(t, \mathbf{x}) \mathbf{R}(\mathbf{x} - \mathbf{a}_\kappa).} \quad (37)$$

It vanishes in the freestream and, near each shock ray, rotates that shock about its edge from the M_0 angle toward the M_1 angle. Because the localization renders $\nabla \cdot \boldsymbol{\beta} \neq 0$, the field enters the bridge through the conservative operator $\mathcal{Q}^\dagger$ of (23), retaining the reaction term of Proposition 1.